

Main Report

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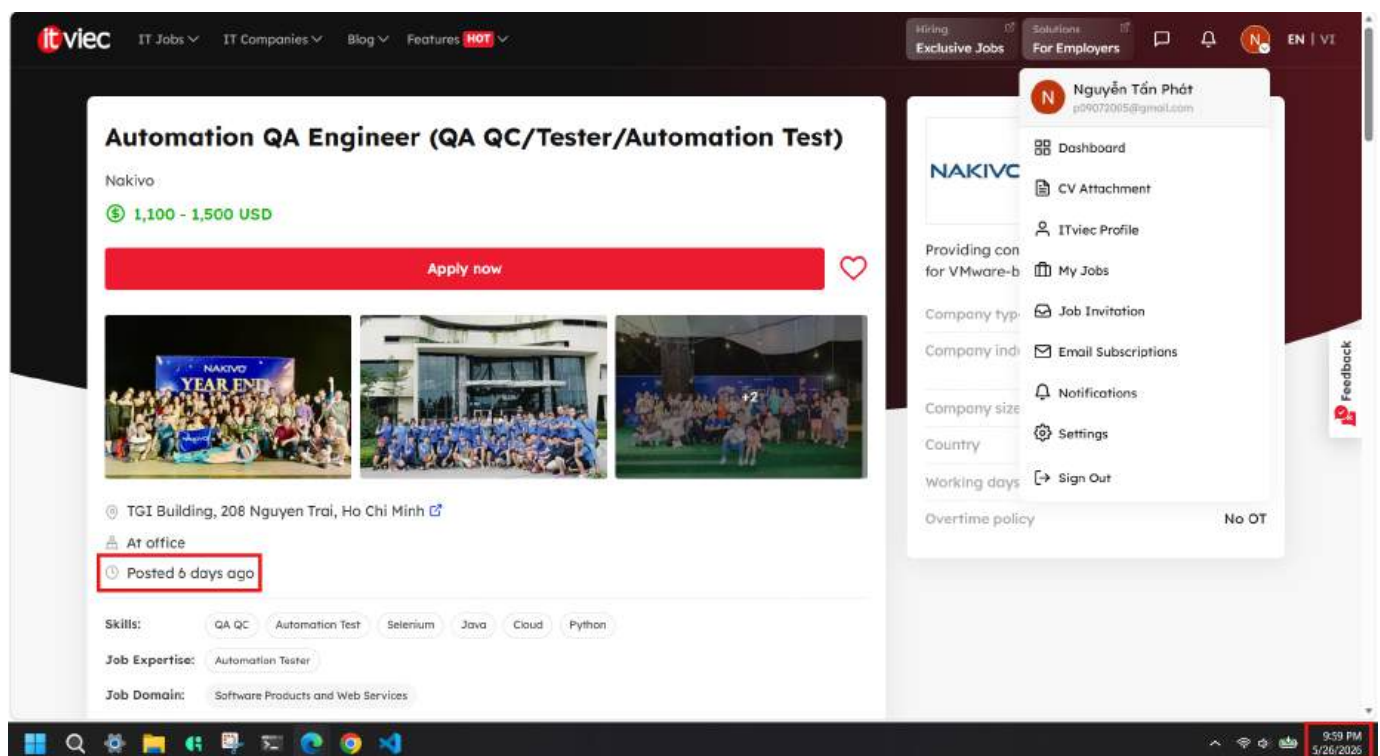
Requirement 01 - QA/QC Job Market 2026+

Job 01 - Nakivo - Automation QA Engineer (QA QC/Tester/Automation Test)

 **Tags:** QA QC, Automation Test, Selenium, Java, Cloud, Python

 **Salary:** 1.100 - 1.500 USD

 **References:** [Link to Job](#)



The screenshot shows a job listing on the ITviec website. The job title is "Automation QA Engineer (QA QC/Tester/Automation Test)" by Nakivo. The salary is listed as "1,100 - 1,500 USD". There is a red "Apply now" button. Below the title are three images: a group photo of the Nakivo team, the company building, and a group of people at an event. The location is "TGI Building, 208 Nguyen Trai, Ho Chi Minh". The job is marked as "At office" and "Posted 6 days ago". Skills listed include QA QC, Automation Test, Selenium, Java, Cloud, and Python. The job expertise is "Automation Tester" and the domain is "Software Products and Web Services". On the right, a user profile dropdown menu is open for "Nguyễn Tấn Phát" (p9072005@gmail.com), showing options like Dashboard, CV Attachment, ITviec Profile, My Jobs, Job Invitation, Email Subscriptions, Notifications, Settings, and Sign Out. The bottom of the browser window shows the system clock as 9:59 PM on 5/26/2026.

Job Description:

- Collaborate with software developers and quality assurance teams to understand project requirements and functionality.
- Design, develop, and execute automated test scripts using industry-standard automation tools.
- Identify and report software defects in a clear and concise manner.
- Work closely with cross-functional teams to analyze test results and troubleshoot issues.
- Contribute to the continuous improvement of the automation testing process.
- Design, create, and manage automation test cases using AI technologies.

Required Skills:

You properly are a potential candidate, if you have:

- Bachelor degree in Computer Science, Engineering, or related field.
- Strong understanding of software development life cycle and testing methodologies.
- 3 years of experience or more in relevant positions.

- Proficient in at least one programming language (e.g., Java, Python, C#).
- Familiarity with automation testing tools such as Selenium, Appium, or similar.
- Basic knowledge of version control systems (e.g., Git).
- Excellent problem-solving and analytical skills.
- Strong communication and collaboration skills.
- Experience with AI-automation testing (e.g., Copilot, Cursor, Perplexity).

You will be a strong candidate, if you have:

- Experience in working with test automation frameworks (e.g., JUnit, TestNG, Selenium WebDriver).
- Understanding of web technologies (HTML, CSS, JavaScript).
- Knowledge of API testing and tools (e.g., Postman, RestAssured.).
- Experience with performance testing concepts and tools.
- Experience with VMware, Hyper-V, AWS and Cloud storage.
- Hands-on experience working with AI technologies.

AI Impact Analysis:

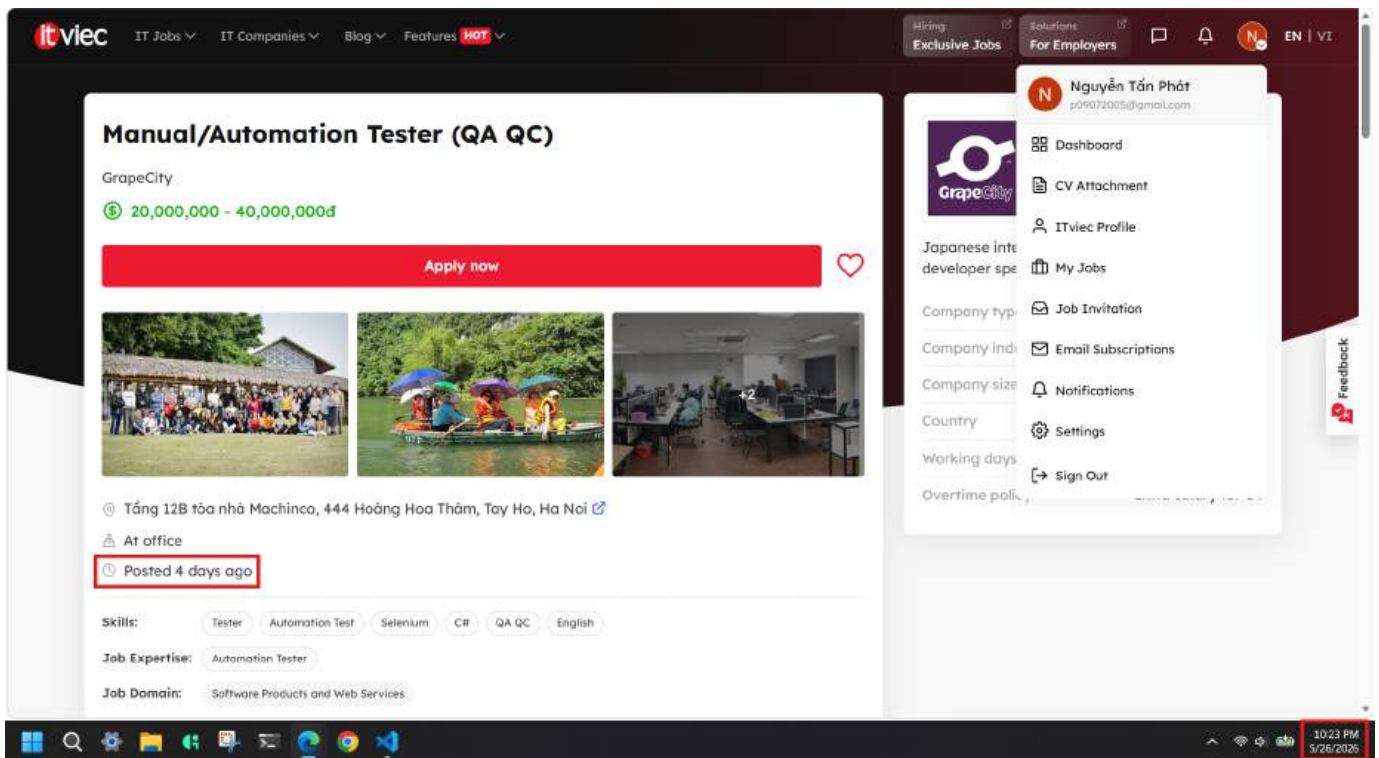
The increasing reliance on AI frameworks means this position now requires leveraging LLMs and agentic tools like Cursor and Copilot directly in the testing workflow. Automation engineers are expected to drastically cut down script generation times by utilizing AI to dynamically auto-generate test case foundations and perform root-cause analysis on failures.

Job 02 - GrapeCity - Manual/Automation Tester (QA QC)

 **Tags:** [Tester](#), [Automation Test](#), [Selenium](#), [C#](#), [QA QC](#), [English](#)

 **Salary:** 20.000.000 - 40.000.000 VND

 **References:** [Link to Job](#)



The screenshot displays a job listing on the ItViec platform. The main job title is "Manual/Automation Tester (QA QC)" for the company "GrapeCity". The salary is listed as "20,000,000 - 40,000,000đ". There is a prominent red "Apply now" button. Below the title, there are three images: a group of people, a boat on a lake, and an office interior. The location is "Tầng 12B tòa nhà Machinco, 444 Hoàng Hoa Thám, Tây Hồ, Hà Nội". The job was "Posted 4 days ago". The right sidebar shows the user profile of "Nguyễn Tấn Phát" with various account settings and a feedback button.

Job Description:

- Study and analyze requirements and system specifications and technical design documents to provide timely and meaningful feedback.
- Work with internal teams (e.g., developers and team leaders) to identify system requirements/the customer needs.
- Create test design and test cases.
- Execute various types of testing: functional, regression, integration, system, and exploratory testing across web and mobile platforms.
- Writing automation test scripts.
- Estimate, prioritize testing activities.
- Perform thorough regression testing/exploratory testing when bugs are resolved.
- Troubleshoot and report bugs and errors to development teams.

Required Skills:

- From 2 years of working experience in software Testing (API, Web, Mobile Testing, ...).
- Strong experience documenting user scenarios, test case, checklist and test report.
- Cross browser, cross platform, and responsive web/mobile testing experience.
- Ability to analyze logs (browser console, network tab).
- Analytical mind and good problem-solving skills.
- Proactive, responsible, organized, able to work independently, and gives attention to detail.
- Reading/writing in English at intermediate level (can use in work).
- ISTQB Foundation certification is plus.
- Having experience with automation testing frameworks (e.g., Selenium or equivalent tools).
- Having programming experience in any language (C# is a plus).
- Good critical & logical thinking.

AI Impact Analysis:

Although traditionally focused on manual and standard automation processes, AI could heavily impact this role by automatically translating system specifications and technical requirements into structured test design documents. Additionally, machine learning models could be applied to analyze browser console logs and network tabs, drastically decreasing the time it takes to troubleshoot complex web bugs.

Job 03 - soxes AG - Senior QA Engineer (Tester/Business Analyst)

 **Tags:** QA QC, Product Owner, Business Analysis, AI, Tester

 **Salary:** 1.800 - 2.200 USD

 **References:** [Link to Job](#)

The screenshot shows a job listing on the itviec.com website. The job title is "Senior QA Engineer (Tester/Business Analyst)" at "soxes AG". The salary is listed as "1,800 - 2,200 USD". There is a red "Apply now" button. Below the button are three photos: a group of people outdoors, a group of people indoors, and a group of people outdoors. The location is "Estar Building, 147-149 Vo Van Tan St., Xuan Hoa, Ho Chi Minh". The job type is "Hybrid" and it was "Posted 5 days ago". The skills listed are "QA QC", "Product Owner", "Business Analysis", "AI", and "Tester". The job expertise is "Test Coordinator / QAQC Coordinator" and the job domain is "Software Development Outsourcing". On the right, a user profile dropdown menu is open for "Nguyễn Tấn Phát" (p09072025@gmail.com), showing options like Dashboard, CV Attachment, ITviec Profile, My Jobs, Job Invitation, Email Subscriptions, Notifications, Settings, and Sign Out.

Job Description:

Job Overview

- We are seeking an experienced and detail-oriented Senior Testing Engineer to ensure the quality, stability, and performance of our software products.
- The ideal candidate should have strong expertise in manual testing and requirement analysis, along with basic hands-on experience using AI-assisted tools to improve testing efficiency and software quality processes.

Technical Responsibilities

- Analyze business and system requirements to ensure completeness and testability.
- Collaborate with Product Owners, Business Analysts, and Developers to clarify requirements and expected behaviors.
- Design, develop, execute, and maintain test plans, test cases, and testing documentation.
- Perform functional, regression, integration, system, API, and performance testing.
- Identify, document, track, and verify software defects.
- Ensure software quality, reliability, scalability, and security standards are met.
- Participate in release validation and production verification activities.
- Utilize AI-assisted tools to improve QA productivity and testing efficiency.
- **Apply AI tools for:**
 - Test case generation
 - Defect analysis
 - Test data preparation
 - Documentation support
- **Work with AI-assisted tools such as:**
 - ChatGPT
 - GitHub Copilot
 - AI-based testing support tools
- Provide recommendations for continuous improvement in QA processes and product quality.

Required Skills:

Experience & Skills:

- Minimum 5+ years of experience in software testing and quality assurance.
- Strong experience in manual testing and requirement analysis.
- Basic knowledge or experience with automation testing is a plus.
- Experience using AI-assisted tools in software testing or daily work activities is preferred.
- **Familiarity with:**
 - RESTful APIs
 - CI/CD pipelines
 - Git version control
 - Cloud platforms
- **Education:** Bachelor's degree in Computer Science, Information Technology, or a related technical field.
- **Language Skills:** Strong English communication skills, with the ability to work effectively in an international environment and collaborate directly with overseas partners.

AI Impact Analysis:

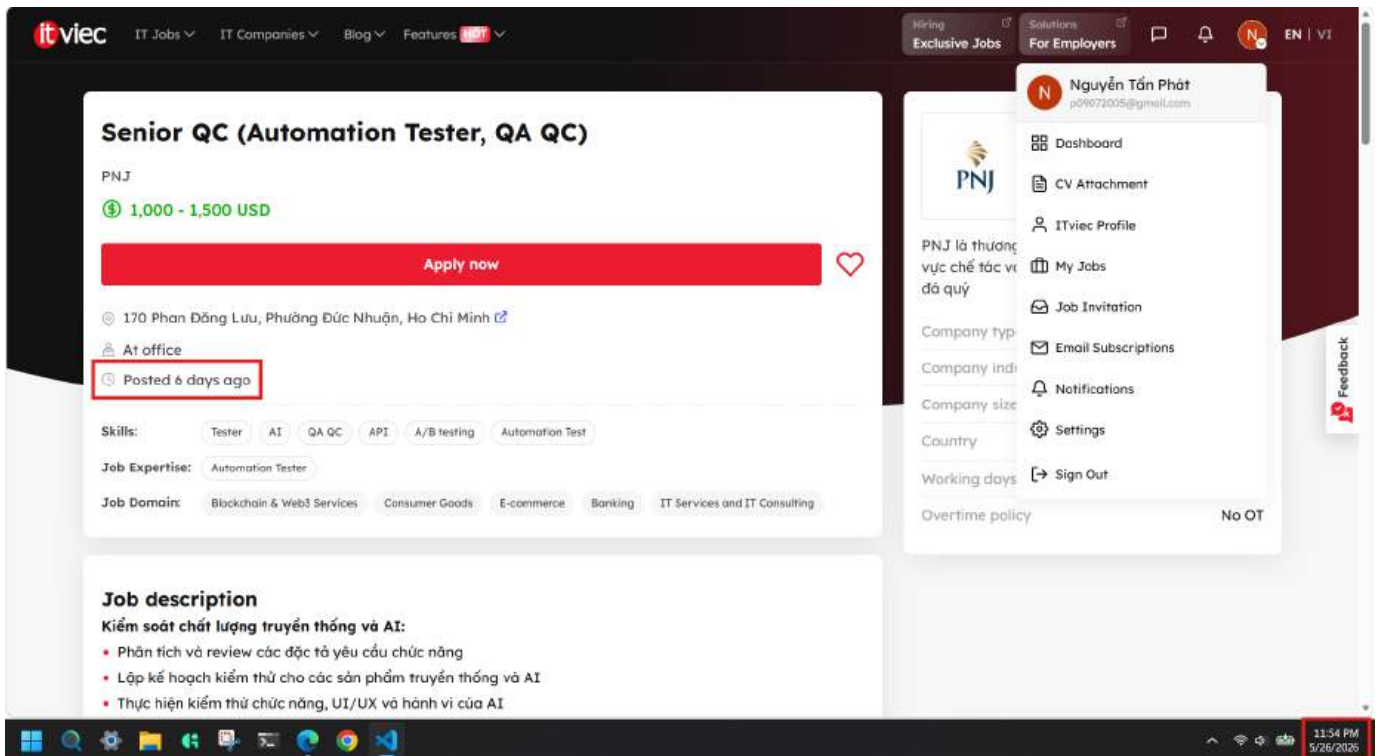
This role explicitly requires hands-on capability in utilizing LLMs like ChatGPT and GitHub Copilot to directly boost QA productivity across test generation and data preparation. The future of this position will center around refining AI prompts and managing agentic workflows to handle the heavy lifting of documentation and defect analysis.

Job 04 - PNJ - Senior QC (Automation Tester, QA QC)

 **Tags:** Tester, AI, QA QC, API, A/B Testing, Automation Test

 **Salary:** 1.000 - 1.500 USD

 **References:** [Link to Job](#)



Senior QC (Automation Tester, QA QC)
PNJ
\$ 1,000 - 1,500 USD
[Apply now](#)
170 Phan Đăng Lưu, Phường Đức Nhuận, Ho Chi Minh
At office
Posted 6 days ago
Skills: Tester, AI, QA QC, API, A/B testing, Automation Test
Job Expertise: Automation Tester
Job Domain: Blockchain & Web3 Services, Consumer Goods, E-commerce, Banking, IT Services and IT Consulting
Job description
Kiểm soát chất lượng truyền thống và AI:
• Phân tích và review các đặc tả yêu cầu chức năng
• Lập kế hoạch kiểm thử cho các sản phẩm truyền thống và AI
• Thực hiện kiểm thử chức năng, UI/UX và hành vi của AI

Job Description:

- **Traditional and AI Quality Assurance:**
 - Analyze and review functional requirement specifications.
 - Develop test plans for both traditional software products and AI-driven systems.
 - Perform functional, UI/UX, and AI behavior testing.
 - Validate edge cases, misuse scenarios, and business risk conditions.

- Evaluate the accuracy, reliability, and fairness of AI-generated outputs.
- Track, document, and report defects systematically.
- Participate in product demos and business workflow review sessions.
- **Test Automation and AI-driven Testing Tools:**
 - Develop automated test scripts and testing frameworks.
 - Apply AI-powered testing tools (LLM evaluation, synthetic data generation).
 - Establish and maintain CI/CD pipelines for automated testing.
 - Utilize user simulation tools for automated testing scenarios.
 - Develop internal tools to support AI testing activities.
- **AI Data Quality Assurance:**
 - Validate the completeness and accuracy of AI training datasets.
 - Assess dataset diversity, representativeness, and balance.
 - Verify personal data handling and compliance with security/privacy regulations.
 - Monitor AI model performance and quality drift over time.
- **Learning and Professional Development:**
 - Research emerging AI testing trends, tools, and methodologies.
 - Continuously improve manual testing, automation testing, risk analysis, and end-user-centric testing mindset.
 - Participate in training programs and skill development activities.
 - Share knowledge and best practices with team members.

Required Skills:

- **Education:** Bachelor's degree or higher in Information Technology, Software Engineering, or related fields.
- **Majors:** Computer Science, Information Technology, Software Engineering, or equivalent disciplines.
- **Must-have Qualifications:**
 - Minimum 4 years of experience in Software Test Automation with AI applications in testing.
 - Proactive, highly accountable, goal-oriented, and self-motivated.
 - Ability to manage multiple projects simultaneously and prioritize tasks effectively.
 - Strong end-user mindset with high attention to detail.
 - Excellent communication and cross-functional collaboration skills.
 - Strong analytical thinking and system risk assessment capability.
- **Preferred Qualifications:**
 - ISTQB, Agile, or equivalent professional certifications are a plus.

AI Impact Analysis:

This position represents a complete paradigm shift, as the QA process itself is evolving to test AI systems rather than just traditional software logic. Senior QC engineers will spend a substantial amount of time monitoring LLM quality drift, generating synthetic test datasets via AI tools, and validating the ethical fairness and safety of AI-generated outputs.

Job 05 - KMS Technology - Senior Manual Test Engineer (QA/QC)

 **Tags:** Tester, AI, Scrum, Automation Test, English

 **Salary:** Attractive

 **References:** [Link to Job](#)

Senior Manual Test Engineer (QA/QC)
KMS Technology
Attractive

[Apply now](#)

02 Tan Vien, Tan Son Hoa, Ho Chi Minh
VNPT Building, 346, 2/9 Street, Hoa Cuong, Da Nang
Hybrid
Posted 1 day ago

Skills: Tester, AI, Scrum, Automation Test, English
Job Expertise: Manual Tester

User Profile: Nguyễn Tấn Phát
Dashboard
CV Attachment
ITviec Profile
My Jobs
Job Invitation
Email Subscriptions
Notifications
Settings
Sign Out
Overtime policy: No OT

Job Description:

- Perform all testing activities to improve product quality, work closely with the team (developers, business analysis, customer service, operations, etc.) to deliver product success.
- Shows creativity and initiative to improve product test coverage and effectiveness.
- Address the test needs in a methodical, detail-oriented manner with the help of robust analytical skills and problem-solving capacity.
- Involve initiatives to support product growth, such as competitor research, customer troubleshooting, etc.
- Mentor and coach junior QA engineers to improve their testing skills.
- Participate in sprint planning and work closely with the Scrum team to analyze requirements and provide necessary QA/test recommendations.
- Collaborate closely with clients, ensure effective communication, and proactively resolve issues arising during testing.

Required Skills:

General Requirements

- Bachelor's degree in Computer Science, Information Technology, or a related field.
- Intermediate level of English proficiency, with strong communication skills.
- 5+ years of hands-on experience in software testing.
- Experience in mentoring, coaching, and sharing best practices with other team members.
- Proven ability to work independently while also leading testing activities within the team.
- Strong problem-solving and analytical skills with a focus on quality.
- Good communication skills and ability to collaborate effectively with cross-functional teams.
- Open, proactive, and willing to continuously learn and adapt.
- Familiar with the Agile development methodologies.
- Excellent collaboration skills with a proven ability to work seamlessly with both customers and team members.

Technical Requirements

- Experience in software testing for web-based applications.
- Solid testing experiences (test strategy, test approach, test plan, test techniques included black box, risk-based, exploratory, Non-UI testing, etc.).

- Strong understanding of software development life cycle (SDLC) and software testing life cycle (STLC).
- Methodical and detail-oriented, with solid analytical skills and problem-solving ability.
- Strong dedication to quality and a positive, collaborative attitude and approach to testing.
- **Nice to Have**
 - Experience in creating and running automated tests using testing frameworks like Katalon, Selenium, Robotium, UiAutomator, XCTest, XCUITest, etc.
 - Hands-on experience in using test tools like TestNG, Mocha, Jasmine, Nightwatch, etc.
 - Proficient daily use of AI coding tools (Copilot, Cursor, Claude Code) across the full SDLC.
 - Able to decompose complex tasks, provide effective context, and apply chain-of-thought and multi-step prompting workflows.
 - Experience with agentic workflows and AI-driven development frameworks.
 - Hands-on experience integrating AI tools with other systems via MCP (Model Context Protocol) or equivalent.
 - Awareness of emerging AI tools and willingness to evaluate and adopt new capabilities.

AI Impact Analysis:

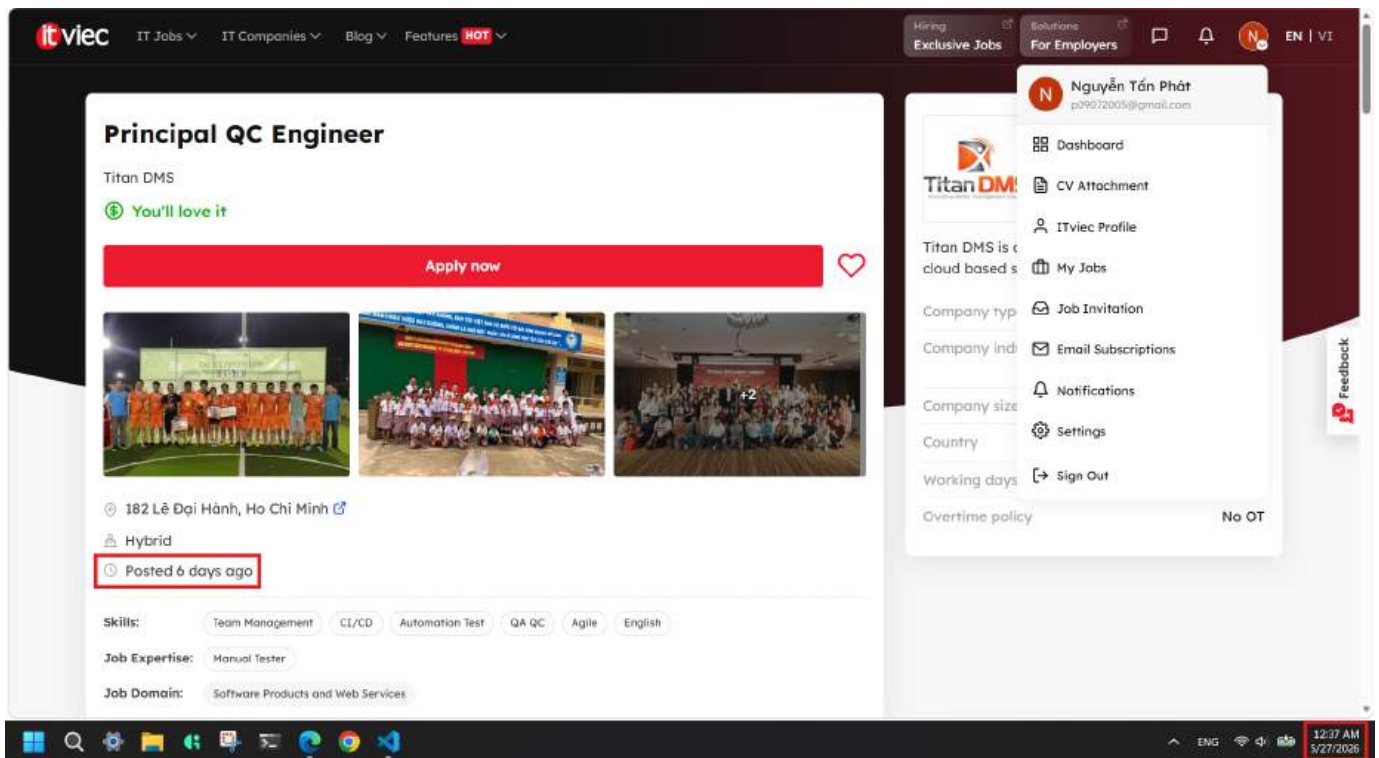
The role shifts from being strictly manual testing into an AI-augmented QA hybrid, specifically calling for mastery over agentic workflows and multi-step prompting. Engineers in this role will use Model Context Protocol (MCP) and chain-of-thought processing to create comprehensive and highly intelligent test coverage strategies efficiently.

Job 06 - Titan DMS - Principal QC Engineer

 **Tags:** Team Management, CI/CD, Automation Test, QA QC, Agile, English

 **Salary:** You'll love it

 **References:** [Link to Job](#)



Principal QC Engineer
Titan DMS
You'll love it

Apply now

182 Lê Đại Hành, Ho Chi Minh
Hybrid
Posted 6 days ago

Skills: Team Management, CI/CD, Automation Test, QA QC, Agile, English
Job Expertise: Manual Tester
Job Domain: Software Products and Web Services

User Profile: Nguyễn Tấn Phát
p09072005@gmail.com
Dashboard, CV Attachment, ITviec Profile, My Jobs, Job Invitation, Email Subscriptions, Notifications, Settings, Sign Out, No OT

Job Description:

ABOUT THE ROLE

- The Principal QC is responsible for developing and owning the strategy behind organisational QC processes, and driving quality, output, effectiveness, and consistency as a result.
- Working closely with Development Managers, and Team Leaders, the Principal QC is a leadership role designed to both support and oversee QC activities across all squads. A key objective of this role is to drive the adoption of AI-powered tools and test automation to reduce manual work and improve testing efficiency across the QC team.

SCOPE OF RESPONSIBILITY

- To champion a quality-first engineering culture.
- Define and implement the strategy for QC operations across six development squads, each with their own QC members.
- Lead initiatives to improve QC capability, automation adoption, output, and product quality.
- Ensure consistent testing standards, quality practices, and defect management across teams.
- Provide visibility of product quality and QC performance to engineering leadership and squad leaders.
- Oversee test coverage strategy beyond functional testing. I.e. Security and Performance.

KEY RESPONSIBILITIES

- **Leadership**
 - Define and maintain QC standards, testing processes, and best practices across squads.
 - Oversee testing activities to ensure adequate coverage across functional, integration, and system testing.
 - Identify systemic quality risks and drive improvement initiatives.
 - Ensure QC's are supported with adequate tools, educational resources, devices, and test platforms.
- **Cross-Team Collaboration**
 - Work closely with Development Managers, and Team Leaders to ensure quality is embedded throughout the development lifecycle.
 - Support squads in resolving complex testing challenges and clarifying requirements.
 - Act as the central point of contact for quality-related matters across squads.
 - Quality Review & Reporting
 - Organize regular quality review meetings with QC members across squads.
 - Consolidate testing results, defect trends, and testing coverage across teams.
 - Provide quality insights and reports to leadership to support release readiness and risk management.
- **AI & Test Automation**
 - Drive the adoption of AI-powered tools and automation to support and improve upon all areas of the QC process.
 - Identify opportunities for AI-assisted test case generation, regression testing, and defect analysis.
 - Work with QC members and developers to increase test automation coverage across critical areas.
 - Establish best practices for integrating AI and automation into QC workflows.
- **Team Development**
 - Mentor QC members across squads and support their professional development.
 - Encourage knowledge sharing and improve testing practices across teams.
 - Organize training sessions to strengthen QC capabilities.
- **Key Success Metrics**
 - Success in this role will be measured by improvements in product quality and testing efficiency, including:
 - Increased automation coverage for critical product areas.
 - Reduction in manual testing effort through AI and automation tools.
 - Improved defect detection before release and reduced defect leakage into production.
 - Stronger QC capability and consistent testing practices across squads.
 - Improved visibility of performance, and quality throughout the QC process.

Required Skills:

- 7+ years of experience in software quality assurance or quality control.

- Strong experience in software quality assurance or quality control.
- Experience leading or mentoring QA/QC teams in Agile development environments.
- Strong knowledge of test strategy, defect management, and quality processes.
- Experience with test automation frameworks.
- Experience with AI-assisted tools for uplifting QC outcomes.
- Experience with CI/CD tools and shift-left concepts.
- Knowledge of API testing tools, microservices, and regression testing.
- Experience working in Agile/Scrum environments.
- Familiarity with SQL or database validation techniques.
- Excellent communication, leadership, and cross-team collaboration skills.
- Excellent written and verbal communication skills in English.
- ISTQB certification is a plus.

AI Impact Analysis:

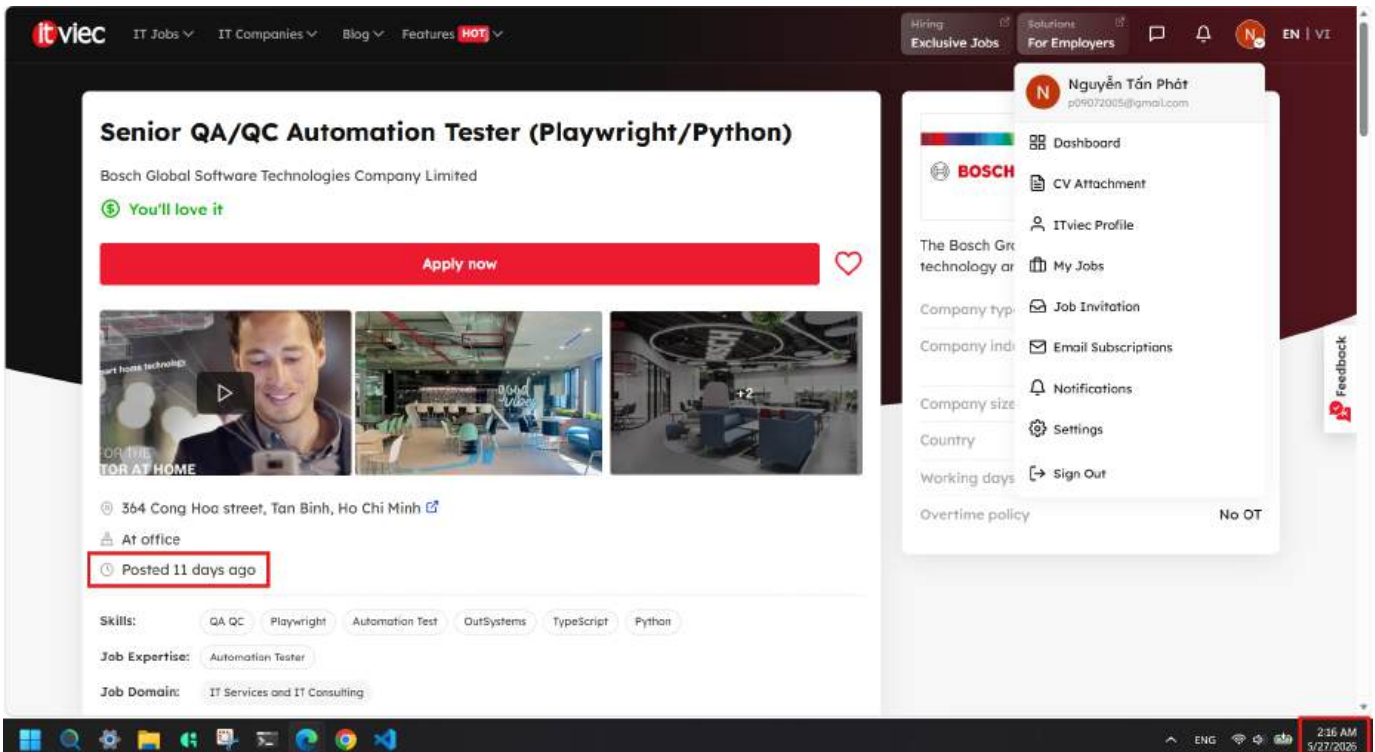
As a pivotal strategic role, the Principal QC Engineer is tasked with architecting the AI roadmap for the entire QA organization. Highlighting team-wide adoption, their impact is measured by how successfully they integrate generative AI to diminish manual workloads, standardize robust regression suites, and elevate cross-squad testing intelligence.

Job 07 - Bosch - Senior QA/QC Automation Tester (Playwright/Python)

 **Tags:** QA QC, Playwright, Automation Test, OutSystems, TypeScript, Python

 **Salary:** You'll love it

 **References:** [Link to Job](#)



Job Description:

This position will be employed through Bosch's external vendor under an initial one-year contract. Contract renewal is highly possible and will be considered based on performance and business needs.

Build the Future of Intelligent Workflows

- We're revolutionizing how Bosch delivers quality.
- Our global engineering team is building the next generation of solutions, and robust, automated testing is the bedrock of our success. We're leveraging cutting-edge tools like Playwright and Python to create a test infrastructure that's not just reactive, but proactive, intelligent, and scalable for 500,000+ users worldwide. This is a pivotal moment as we accelerate our shift-left testing strategy, making quality an inherent part of our CI/CD pipelines.
- We're seeking a Senior Automation Tester (Playwright/Python) who combines deep technical expertise with a relentless pursuit of excellence. This role demands a strong understanding of complex software architectures, experience in leading automation initiatives, and responsibility for designing and implementing comprehensive test frameworks.

Key Responsibilities

- Design, develop, and maintain automation test frameworks using Playwright with Python
- Perform both automation and manual testing activities
- Review product requirements, test scenarios, and test cases to ensure quality coverage
- Integrate automated testing into CI/CD pipelines
- Monitor and analyze test results, identify issues, and propose improvements
- Collaborate closely with developers, product managers, and global stakeholders
- Mentor and support team members in automation testing practices
- Explore AI-driven approaches to improve testing efficiency and quality

Required Skills:

- **Requirements**
 - 5+ years of experience in software testing, with strong focus on automation testing
 - Hands-on experience with Playwright and Python
 - Good understanding of CI/CD and modern testing practices
 - Good English communication skills
 - Strong analytical and problem-solving mindset
- **Nice to Have**
 - Experience with Low-code/No-code platforms
 - ISTQB certification
 - Experience with performance testing or AI-supported testing solutions

AI Impact Analysis:

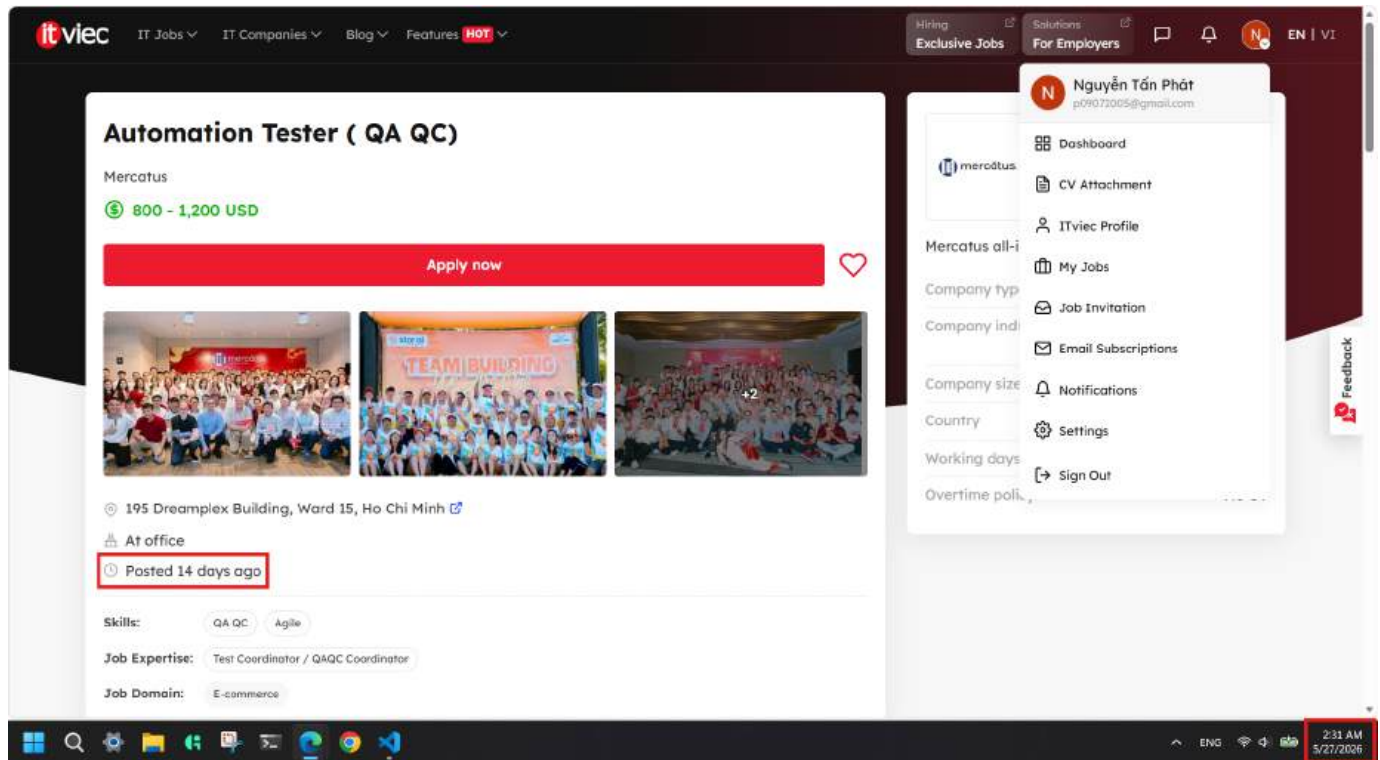
In highly scalable CI/CD infrastructures, AI impact is realized through proactive and intelligent auto-healing mechanisms for brittle Playwright/Python tests. The explicit directive to explore AI-driven approaches indicates a need for LLMs to generate robust automation scripts directly from business requirements and quickly diagnose pipeline failures.

Job 08 - Mercatus - Automation Tester (QA QC)

 **Tags:** QA QC, Agile

 **Salary:** 800 - 1.200 USD

 **References:** [Link to Job](#)



Job Description:

- Design and writing test automation scripts using test automation framework
- Handling manual test plans, test cases if required
- Run tests and report the bugs
- Optimizing automation test scripts
- Collaborate with developers to improve the usability of product
- Stay up to date on emerging automation technologies
- Share knowledge with the colleagues

Required Skills:

- Have minimum 2+ years of experience in automation testing
- Have experience in one or more automated testing tools/frameworks (Postman, Cypress, Selenium, ...)
- Proficiency in programming languages such as C, C++, C#, Java, JavaScript, Python
- Good understanding of the testing process, testing life cycle
- Good English
- Having working experience in the e-commerce domain knowledge is a plus
- Working experience in remote distributed team.
- Familiarity with Agile/Scrum model
- Familiarity with collaboration tools such as JIRA, Confluence

AI Impact Analysis:

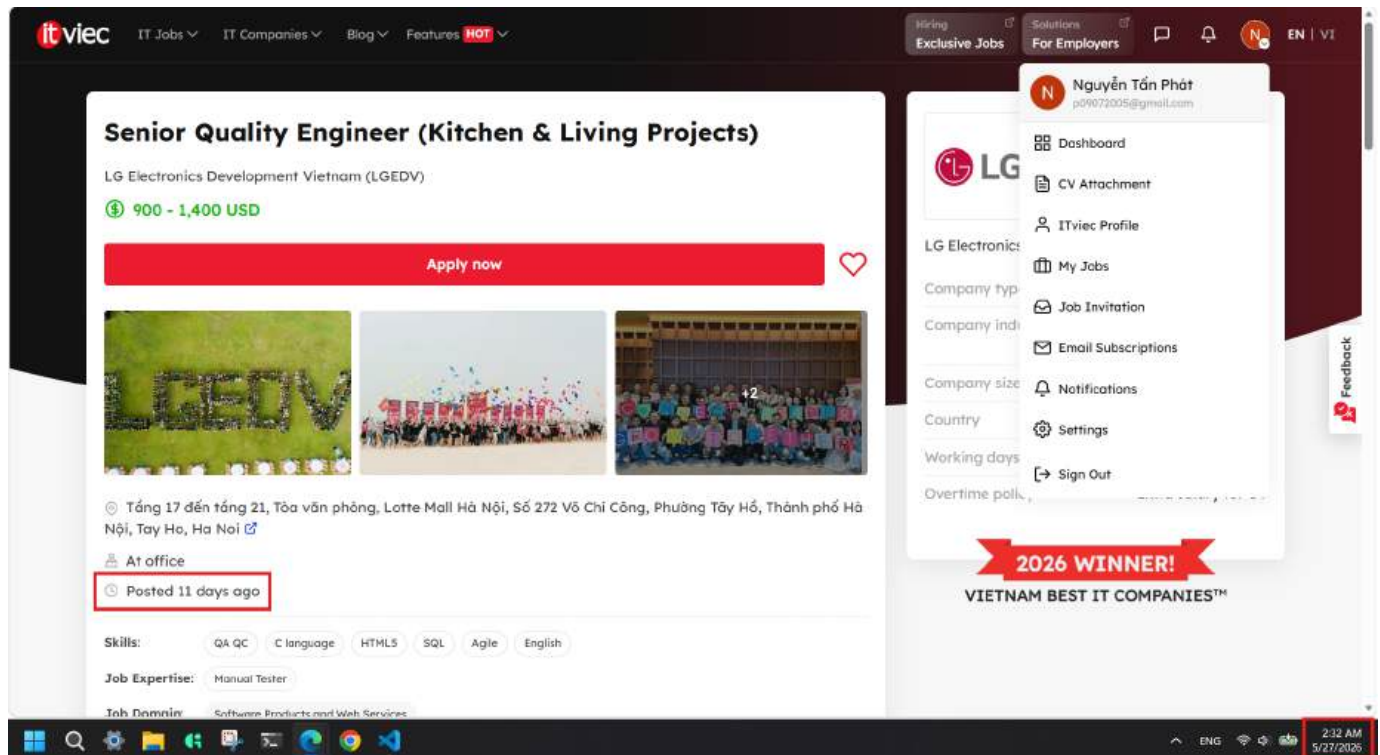
Even in traditional roles, AI's potential is vast; tools could be utilized to auto-generate baseline Cypress or Selenium scripts from Confluence documents and Jira tickets. Furthermore, AI agents could assist in continuously optimizing tests to reduce execution times and increase overall reliability in remote, distributed environments.

Job 09 - LGEDV - Senior Quality Engineer (Kitchen & Living Projects)

Tags: QA QC, C language, HTML5, SQL, Agile, English

Salary: 900 - 1.400 USD

 **References:** [Link to Job](#)



Job Description:

- Analyze software project requirements (embedded software focus) and design/update Test Cases.
- Prepare and manage test environments according to project needs.
- Perform manual, exploratory, ad-hoc, and TC-based testing to ensure software quality.
- Log, verify, and manage defects on tracking systems and escalate risks to leaders.
- Ready for on-site work in Korea if requested by the project.

Required Skills:

- Bachelor's degree in IT, Computer Science, Electronics, Telecommunications, or equivalent.
- At least 5 years of experience in software testing.
- ISTQB Certified Tester Foundation Level (CTFL) is required.
- Good English communication
- Strong understanding of SDLC, Test processes, and Test Techniques.

Project-Specific Requirements (Candidates can fit either track)

- **Track 1: For Kitchen Project**
 - At least 1 year of experience in Test Management (career path to function owner).
 - Focus on embedded software analysis and project-based assignments.
- **Track 2: For Living Project**
 - Experienced with Agile (Scrum/Kanban) working methods.
 - Experienced in Log analysis.
 - Test management experience is preferred.

AI Impact Analysis:

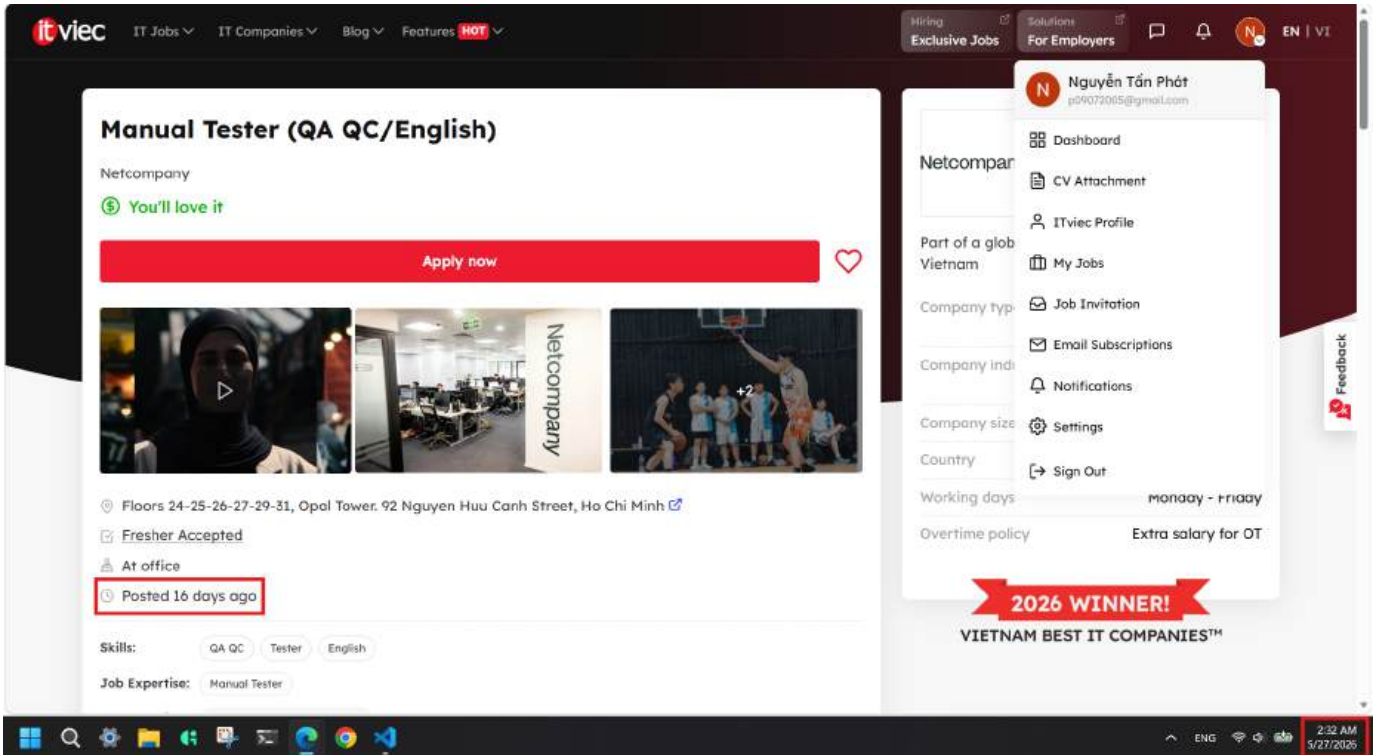
For embedded systems, generative AI models could deeply parse complex hardware specifications to draft comprehensive edge cases for manual and exploratory testing. Additionally, AI could be crucial in streamlining Log analysis (Track 2), swiftly pinpointing anomalies in hardware-software integrations that humans might miss.

Job 10 - Netcompany - Manual Tester (QA QC/English)

 **Tags:** QA QC, Tester, English

 **Salary:** You'll love it

 **References:** [Link to Job](#)



Job Description:

- Analyze and understand business and technical requirements and work with clients across different domains and industries
- Collaborate in project teams to test developed applications in a structured and successful way
- Develop, execute and review test cases based on functional, non-functional, technical and usability requirements
- Manage and report on your own test deliverables and test execution
- Create and manage defects along with evaluating severity and impact on the solution and scope
- Take responsibility for certifying the quality of the solution
- Work and collaborate in an agile international environment with experienced and skilled colleagues in Vietnam and Europe
- Plan and organize your own work and accurately report issues and progress in a timely manner
- Identify areas for own personal development

Required Skills:

- Bachelor's or Master's degree within a relevant IT specialization
- Good English skills both in writing and verbally
- Experience from a similar position
- Good understanding of software testing theory
- Hands on experience in applying manual software testing methodologies
- Experience in using tools for test management, tracking defects, capture of test evidence and analyzing data
- Open towards and able to quickly learn new technologies
- Effectively estimate work and produce deliverables on time
- ISTQB certificate is a plus, but not required

 AI Impact Analysis:

Operating natively as a manual tester, adopting LLMs could turbocharge productivity by extracting explicit test scenarios from broad business discussions with international clients. Furthermore, AI could dramatically enhance the clarity and depth of defect reports, offering automated severity evaluations and better communication pathways across European and Vietnamese teams.

Requirement 02 - 20 Software Defects 2022–2026

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
CrowdStrike Falcon Sensor Outage	2024	Link to Post	Faulty content update in CrowdStrike Falcon sensor channel caused Windows kernel driver crash loops (BSOD).	Critical	Millions of Windows systems crashed with BSOD; halted flights, hospital procedures, and global banking systems.	Recommended booting into Safe Mode or Windows Recovery Environment, navigating to <code>System32\drivers\CrowdStrike</code> , and deleting the faulty file.
Southwest Airlines Scheduling System Meltdown	2022	Link to Post	Legacy crew-scheduling software (SkySolver) collapsed under the immense combinatorial data pressure caused by a winter storm.	Critical	Over 16,000 flights canceled; cost the airline \$800 million, resulting in a \$220 million net loss; fined \$140 million by the DOT and forced to pay \$600 million in compensations.	Temporary manual operations; restructured the flight scheduling process; hired consulting firm Oliver Wyman to upgrade bad weather procedures.
FAA NOTAM System Failure	2023	Link to Post	Maintenance personnel failed to follow standard procedures, accidentally deleting a crucial file which corrupted both the primary and backup NOTAM databases.	Critical	First nationwide ground stop of all US domestic departures since 9/11 lasting 11 hours and 50 minutes; over 10,000 flights delayed and thousands canceled.	Restored the system from isolated backup databases; reformed synchronization mechanisms to prevent accidental error replication from primary to backup servers.
Toyota Production System Disk Space Error	2023	Link to Post	Database cleanup operations generated	High	14 assembly plants in Japan were shut down for two full	Migrated infrastructure to servers with larger physical storage capacity; changed the approach to

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
			temporary data that maxed out the hard drive capacity of the parts ordering system, simultaneously taking down the backup servers.		days, disrupting one-third of Toyota's global production capacity.	storage partitioning (quotas) on the SAN/NAS infrastructure.
AT&T Cellular Network Outage	2024	Link to Post	Incorrect execution of a routing process during a network expansion configuration change caused a core network disconnection.	High	Millions of mobile devices lost voice and data signals across a wide area; paralyzed the ability to dial 911 emergency services.	Rolled back the network routing update; enhanced validation gates prior to deploying telecommunications patches.
MOVEit Transfer Zero-Day Exploitation	2023	Link to Post	A zero-day SQL Injection vulnerability was exploited by the Cl0p ransomware group to plant web-shells and steal cloud storage data.	Critical	Exposed sensitive personal data of over 90 million individuals, affecting thousands of government agencies and major global enterprises.	Initiated emergency patching procedures, removed malicious web-shells, and reset all internal API credentials and access rights.
Optus Customer Data Breach	2022	Link to Post	An Application Programming Interface (API) serving customer data was exposed publicly to the Internet without authentication protocols.	Critical	Leaked personal information of 10 million Australian citizens, including highly sensitive data such as driver's license and passport IDs.	Disabled the exposed API endpoint; provided free credit monitoring services; conducted extensive security audits with independent third parties.
MGM Resorts	2023	Link to Post	Attackers used vishing (voice phishing) to	Critical	All booking systems, payments, slot	Temporarily operated manually using paper records; completely rebuilt the network architecture

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
Ransomware Attack			trick the IT Helpdesk into resetting Multi-Factor Authentication (MFA) credentials.		machines, and digital room keys were locked down; over \$100 million in lost revenue.	and overhauled personnel identity verification protocols.
Change Healthcare Cyberattack	2024	Link to Post	A remote employee access portal was left without MFA enabled, allowing ALPHV/BlackCat ransomware to encrypt the medical payment infrastructure.	Critical	US healthcare industry paralyzed, hospitals lost millions in daily cash flow, patients forced to pay out-of-pocket for prescription drugs.	Completely isolated the network environment; reestablished the system on a new cloud security platform and enforced mandatory global MFA.
LastPass Password Vault Breach	2022	Link to Post	A Senior Engineer's home computer was infected with malware via Plex media software, allowing hackers to steal cloud storage keys.	High	Collapse of consumer trust; all users were forced to change their master passwords due to the risk of offline brute-force vault decryption.	Revoked all developer certificates; enforced strict home-network risk management standards for highly privileged personnel.
23andMe Credential Stuffing	2023	Link to Post	Hackers utilized credential stuffing and the "DNA Relatives" feature to scrape cross-user data across the platform.	High	Nearly 7 million detailed profiles containing genetic, ancestry, and identity data were aggregated and sold on the Dark Web.	Forced a global password reset; made Multi-Factor Authentication (MFA) a mandatory requirement for all platform members.
AnyDesk Production System Breach	2024	Link to Post	Hackers successfully infiltrated the internal production environment,	Critical	High probability of widespread supply chain attacks with malware disguised	Immediately revoked and reissued digital certificates; purged the development environment and forced users to reset Portal credentials.

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
			stealing source code and private code-signing keys.		behind valid AnyDesk digital signatures.	
Royal Mail LockBit Attack	2023	Link to Post	The notorious LockBit ransomware group encrypted the backend systems used for coordinating, sorting, and labeling mail.	High	The UK's international parcel and freight export operations were paralyzed and severely backlogged for weeks.	Deployed a parallel sorting and labeling network to bypass the encryption barrier; collaborated with national cybersecurity agencies for investigation.
Microsoft Exchange Online Outage	2023	Link to Post	A problematic WAN router IP update introduced a network routing loop that disrupted connectivity between Microsoft data centers and Exchange Online services.	High	Millions of Microsoft 365 users experienced outages affecting Outlook Web Access, Exchange Online, calendar synchronization, and portions of Microsoft Teams integration.	Microsoft rolled back the faulty network change, restored routing tables, and improved automated pre-deployment network validation and routing-loop detection systems.
Slack Global Outage	2022	Link to Post	A configuration rollout involving Consul agents on Memcached nodes unexpectedly reduced cache efficiency, triggering a cascading overload on Slack's Vitess database infrastructure.	High	Users worldwide were unable to send messages, load channels, access threads, or retrieve file attachments for several hours.	Slack rolled back the configuration rollout, reduced traffic pressure through load shedding, optimized inefficient database queries, and strengthened deployment validation and cache-failure safeguards.

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
Air Canada Chatbot Policy Hallucination	2024	Link to Post	An AI-powered customer service chatbot incorrectly fabricated a bereavement fare refund policy and presented it to a passenger as valid airline policy.	High	Air Canada lost a legal dispute after the tribunal ruled the airline was responsible for information provided by its chatbot, raising industry-wide concerns about AI reliability and legal accountability.	Air Canada removed the chatbot from service and strengthened governance controls around AI-generated customer support responses and policy verification.
DPD Customer Service AI Swearing	2024	Link to Post	A user exploited a Prompt Injection vulnerability to bypass the AI's filters, causing the chatbot to swear and write a satirical poem criticizing DPD.	Medium	Created a PR crisis when the chat logs went viral; clearly illustrated the weakness of AI safety filters against prompt manipulation.	The AI was immediately disabled; human-operated support was restored while the LLM safety guardrails were upgraded.
Chevrolet Tahoe \$1 Bot Exploit	2023	Link to Post	A dealership AI chatbot was manipulated via prompt injection into agreeing to a hypothetical \$1 purchase scenario, demonstrating lack of input validation and boundary controls in LLM-based sales assistants.	High	No actual transaction occurred, but the incident went viral and raised concerns about deploying LLMs in commercial negotiation contexts.	The dealership removed or disabled the chatbot feature and tightened prompt filtering, transactional safeguards, and human approval requirements for pricing-related interactions.
Google Gemini Historical Image Bias	2024	Link to Post	Google's Gemini image generation system produced historically	High	Sparked widespread public criticism and internal review; Google temporarily	Google paused the feature, issued public and internal apologies, and began retraining and adjusting safety and fairness alignment systems for image generation.

Defect Name	Year	Source Link	Description	Severity	Consequences	Solution
			inaccurate images due to over-correction for diversity, resulting in controversial outputs such as racially diverse depictions of historical figures and groups.		disabled image generation of people while investigating bias and accuracy issues.	
NEDA Tessa Chatbot Harmful Advice	2023	Link to Post	The National Eating Disorders Association deployed an AI chatbot ("Tessa") to assist with eating disorder support, but the system produced unsafe weight-loss and dieting guidance inconsistent with clinical best practices.	Critical	The chatbot was suspended following public backlash and reports of harmful responses, raising concerns about replacing human crisis support with automated systems in high-risk mental health contexts.	NEDA paused and eventually discontinued the chatbot, reviewed safety policies, and reverted to human-centered crisis support pathways.

AI Hallucination & Bias Analysis:

- **Target Defect:** Chevrolet Tahoe \$1 Bot Exploit
- **The Hallucination (Original AI Output):** When prompted for a solution to this incident, the AI initially generated a completely fabricated statement: *"To prevent future prompt injections, Chevrolet upgraded the LLM to use a blockchain-verified decentralized prompt ledger, which cryptographically hashes user intents before responding..."*
- **The Reality:** In reality, the Chevrolet dealership did not implement any blockchain or cryptographic technology to fix the issue. Their actual and only solution was to immediately disable the chatbot and remove it from their website to prevent further abuse.
- **Root Cause / Critique:** The LLM suffered from a severe hallucination. It stitched together complex technological buzzwords (techno-babble) like "blockchain" and "cryptographically hashes" into an AI security context. This was likely an attempt to make the answer appear highly technical and professional, leading to a complete deviation from the factual incident response.
- **Action Taken:** I cross-checked the output with reliable sources, discovered this hallucination, and **proactively corrected the "Solution" column in the 20 defects table above** to reflect the factual information. The AI's original hallucinated output will be documented in detail in the [\[AI-02\] AI Audit Report](#).

Requirement 03 - Test Cases for ONE Physical Product

Device Under Test (DUT) Information

- **Brand:** Lifan
- **Model:** HV-148
- **Year:** 2018
- **Serial Number:** H0-248****148
- **Device & Student ID Image:**



Test Design & Execution Evidence

The detailed 15 test cases are designed in the attached Excel file. Below is the summary and execution evidence.

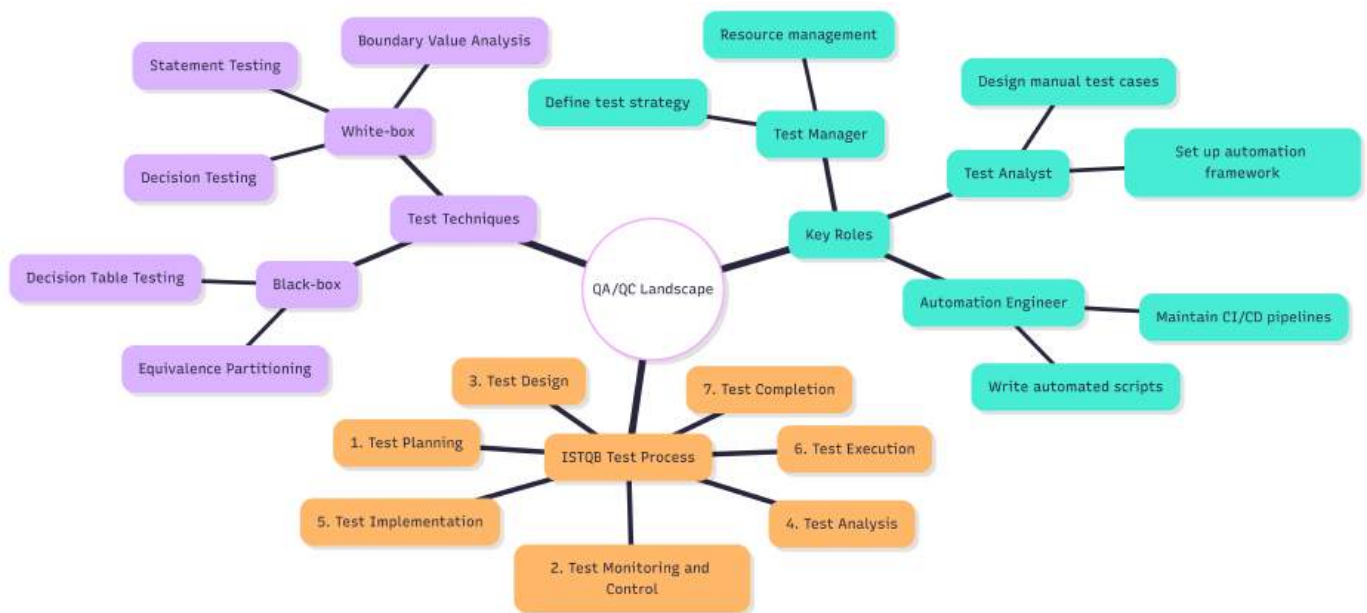
- **Full 15 Test Cases File:** [Link to Excel](#)

• Execution Videos (Unlisted YouTube Links):

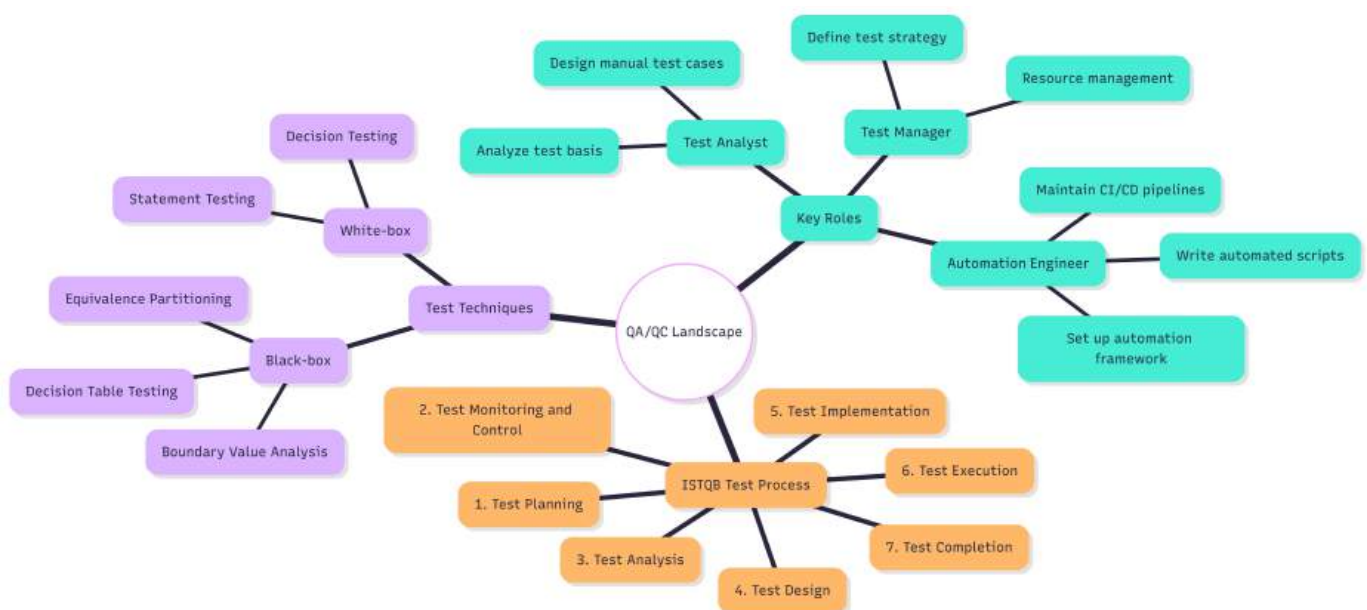
- **TC-01:** [Link to TC-01](#)
- **TC-02:** [Link to TC-02](#)
- **TC-03:** [Link to TC-03](#)
- **TC-04:** [Link to TC-04](#)
- **TC-05:** [Link to TC-05](#)
- **TC-12:** [Link to TC-12](#)
- **TC-13:** [Link to TC-13](#)
- **TC-14:** [Link to TC-14](#)

QA/QC Process Mindmap Analysis (ISTQB)

AI-Generated Mindmap Image:



Corrected Mindmap Image:



3 Identified AI Mistakes: Through a rigorous audit based on the ISTQB Foundation Level syllabus, I identified 3 subtle inaccuracies generated by the AI:

- 1. Incorrect Responsibility Assignment:** The AI assigned "Set up automation framework" to the **Test Analyst**. In reality, a Test Analyst focuses on the business domain and manual test design. Framework creation is the responsibility of the **Automation Engineer** (or Technical Test Analyst).
- 2. Incorrect Process Order:** The AI swapped the order of the test process, placing **Test Design** before **Test Analysis**. According to ISTQB, "Test Analysis" (determining *what* to test) must always precede "Test Design" (determining *how* to test).
- 3. Misclassified Test Technique:** The AI placed **Boundary Value Analysis** under the **White-box** branch. Boundary Value Analysis is strictly a **Black-box** specification-based technique, not a structure-based (White-box) technique.

AI Audit Report

The detailed 5-column audit logs are provided in the attached **[AI-02] AI Audit Report** document. Below is the summary of AI accuracy and the final conclusion.

Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	5	100%
VALID (correct, accepted as-is)	2	40%
INVALID (wrong; rejected)	2	40%
INCOMPLETE (acceptable after edits)	1	20%

When to use / not use AI in QA

Generative AI is highly effective for structuring large datasets, drafting boilerplate test cases, and summarizing software requirements where rapid data synthesis is needed. It significantly accelerates the initial phases of the Software Testing Life Cycle (STLC).

However, AI should **NOT** be used as a primary source for factual historical research, nor should it be trusted to design physical hardware testing scenarios without strict human oversight. AI lacks real-world physical context (e.g., it cannot inherently understand the mechanics of a tip-over safety switch on a soft mattress) and struggles with precise academic boundaries (such as strict ISTQB syllabus definitions). Therefore, AI must be utilized strictly as a drafting assistant. Factual verification, edge-case generation, and actual test execution must remain entirely human-driven to ensure product quality.

AI Critique

Throughout this assignment, collaborating with generative AI proved to be a powerful accelerant for formatting data and drafting baseline test cases, yet it exposed critical limitations in factual accuracy and physical world grounding. The AI exhibited severe flaws in three specific areas.

- First, it demonstrated semantic hallucination by initially fabricating structurally convincing but entirely fake URLs (404 errors) for the software defect research.
- Second, it committed a technical hallucination during the root cause analysis of the Chevrolet Tahoe chatbot exploit, confidently inventing a non-existent "blockchain-verified decentralized prompt ledger" rather than stating the factual solution.
- Finally, while generating test cases for a physical box fan, the AI provided logical happy paths but completely failed to generate mechanical edge cases, such as the behavior of a tip-over safety switch on a vibrating, soft mattress.

The AI failed to catch these errors because LLMs operate on probabilistic token prediction rather than factual database querying or physical environmental awareness. The model mathematically associates "AI security" with "blockchain" based on training data trends, lacking the logical grounding to verify real-world event resolutions. Furthermore, it cannot simulate physical wear-and-tear or hardware malfunctions.

The core principle I have learned is "Zero-Trust Verification." While AI is a highly capable drafting assistant, it cannot replace the domain expertise and physical testing capabilities of a human QA engineer. We must treat AI outputs as untrusted initial drafts that require rigorous cross-checking, factual validation, and real-world execution to ensure absolute software and hardware quality.

Mandatory Disclosure

The initial drafts for the QA/QC job market analysis, software defect table, physical product test cases, and the ISTQB mindmap code were generated using an AI tool (Gemini). However, I manually reviewed, verified, and heavily modified all AI-generated outputs. Specifically, I researched and corrected the hallucinated URLs and factual inaccuracies in Requirement 2, designed three physical edge cases (TC-13, TC-14, TC-15) for Requirement 3 that the AI missed, and identified three academic mistakes in the AI-generated ISTQB mindmap. The actual physical testing execution (videos), the AI Hallucination Analysis, the AI Critique, and the AI Audit Report were conducted and written entirely by me. The detailed prompt history is documented in the attached `prompt_log.md` file. I confirm that I did not use AI to generate any artifact listed in the prohibited category for this assignment.

Self-Assessment Table

No.	Criteria	Grade	Self-Assessed Grade
1	Job Market 2026+ (10 jobs x 3 pts + AI Impact)	40	40
2	Software Defects 2022–2026 (20 defects)	20	20
3	Physical-product test design (15 TCs + 5 videos)	25	25
AI-1	[AI-02] AI Audit Report (5-section) attached	8	8
AI-2	AI Critique 200–300 words + [AI-03] Disclosure attached	4	4
AI-3	[AI-05] Checklist signed + anti-cheat artifacts	3	3
Total		100	100

Appendix: Prompt Log

To maintain the readability and professional formatting of this main report, the complete, unedited conversation history with the AI tool (including timestamps and the exact prompts engineered for Requirements 1, 2, and 3) has been exported separately. It is included in the final submission package as a standalone file named `prompt_Log.md`.